Basu Tandon

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Professional Summary

Data-driven engineer with over 2 years of experience in process optimization, predictive modeling, and analytics at Tata Steel. Skilled in Python (Pandas, NumPy, Scikit-learn), Power BI, and Excel for deriving actionable business insights from complex industrial datasets. Passionate about applying data science and AI to enhance sustainability, efficiency, and decision-making in manufacturing environments.

Academic Profile

Year	Degree	Institution	CGPA/%
2021–2023	M.Tech in Materials Science & Engineering	Indian Institute of Technology, Kanpur	8.75/10
2017–2021	B.Tech in Metallurgical & Materials Engineering	National Institute of Technology, Durgapur	8.28/10
2016	Class XII (CBSE)	Gandhi Nagar Public School, Moradabad	75%
2013	Class X (CBSE)	Gandhi Nagar Public School, Moradabad	9.20/10

Professional Experience

Tata Steel Limited, Technologist – Iron Making Technology Group Mar 2024 – Present

- Analyzed large-scale rotary kiln data using **Python (pandas, numpy, matplotlib)** to correlate process parameters with DRI quality, coal consumption, and temperature distribution.
- Developed and deployed Python-based analytical tools to optimize DRI kiln performance, including a Specific Coal Consumption Analysis and Product Quality (Fe_M) Correlation Study; utilized `oracledb` and `seaborn` to identify key KPIs and establish optimum SAF fan setpoints.
- Created automated reports and **Power BI dashboards** for plant managers to monitor key performance indicators (FeM%, energy efficiency, breakdown risk).
- Implemented a 50ºC reduction in kiln temperature through data-driven modeling, resulting in **30 crore annual savings** and longer refractory life.
- Led the application of **INSULAIRR® Li 1000 coating**, reducing heat loss and saving 1.5 crore annually.
- Authored multiple technical reports, including:
  - Data-Driven Trial of Rice Husk as a Replacement for Coal in DRI Rotary Kiln.
  - Heat and Mass Balance Optimization for Energy Efficiency in Rotary Kilns.
  - Sustainability Evaluation using PKSC in Coal-Based DRI Production.

Tata Steel Limited, Management Trainee – Iron Making Technology Group Aug 2023 – Feb 2024

- Conducted process data analytics and developed models for improving DRI performance; secured 2nd place in Tata Steel MT Innovation Project 2024.

Data Analytics & Modeling Projects

Predictive Modeling of DRI Quality using Process Data 2024

- Applied data preprocessing, correlation analysis, and feature selection on kiln datasets.
- Trained regression models (Linear, Random Forest) using **scikit-learn**, achieving $R^2 = 0.92$ in predicting metallization (FeM%).
- Visualized variable dependencies and model predictions using **matplotlib and seaborn**.

Preparation of Al-Doped ZnO Thin Films & ML-Based Optimization 2022 – 2023

- Optimized thin-film processing parameters using **Genetic Algorithm and Regression Models**.
- Achieved 15% improvement in resistivity prediction accuracy by cross-validation.
- Characterized samples using XRD, SEM, and UV-Vis spectroscopy for validation.

Hardness Variation with Carbon Concentration in Steel 2020 – 2021

- Developed a MATLAB model using Finite Difference Method to simulate hardness-depth variation.
- Visualized and analyzed results to quantify diffusion and mechanical response.

Certifications

- NPTEL: Nanotechnology Science and Applications (IIT Madras)
- Coursera: Structuring Machine Learning Projects – DeepLearning.AI
- Coursera: Neural Networks and Deep Learning – DeepLearning.AI
- A2 Level Certification in German Language (2022)

Technical Skills

- **Data Analysis:** Python (Pandas, NumPy, Matplotlib, Seaborn), Power BI, Excel (Pivot, VLOOKUP, Solver)
- **Machine Learning:** Scikit-learn (Regression, Classification, Feature Engineering)
- **Database:** SQL (Joins, Aggregations, Queries)
- **Other Tools:** MATLAB, C, AI Prompt Engineering, MS Office

Scholastic Achievements

- Awarded DAAD Scholarship, TU Braunschweig, Germany (2022–2023)
- Achieved AIR 420 in GATE Metallurgical Engineering (2021)
- Secured AIR 40799 in JEE Mains (2017) among 1.2 million applicants